

EdgeBench: A Lightweight Preprocessing and Streaming Ingestion Framework for Edge AI Pipelines

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Real-time edge AI workflows require low-overhead preprocessing to handle continuous data streams before routing payloads to high-performance industrial backends. This preliminary work introduces **EdgeBench**, a lightweight streaming ingestion framework engineered to profile system-level telemetry across heterogeneous hardware targets. Built as a modular, bounded-state pipeline, EdgeBench decouples data transport from core analysis to maintain a completely flat physical memory footprint. Initial cross-platform evaluations on an x86 workstation and an embedded ARM-based Xilinx PYNQ platform demonstrate portable execution logic and successfully isolate front-end frame-decoding bottlenecks. This infrastructure establishes a scalable foundation for integrating multi-modal sensor streams within distributed computing pipelines.

Additional Key Words and Phrases: Edge AI, Real-Time Systems, Heterogeneous Computing, Runtime Benchmarking, Performance Evaluation, Stream Processing, Embedded Systems.

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1 Summary

Real-time edge artificial intelligence (AI) workflows increasingly require efficient data preprocessing and ingestion mechanisms to handle continuous data streams from experimental scientific instrumentation, cyber-physical systems, and distributed manufacturing clusters before routing to high-performance computing (HPC) backends [1–7]. While downstream machine learning algorithms excel at complex anomaly detection, deploying these resource-intensive models directly onto heterogeneous edge devices often introduces severe processing bottlenecks and high latency. Furthermore, standard evaluations frequently emphasize algorithmic detection accuracy while providing limited insight into critical system-level runtime performance characteristics, including end-to-end latency, throughput, CPU utilization, and memory consumption, that dictate pipeline stability [4, 8–10]. Developers routinely implement task-specific streaming pipelines tightly coupled to standalone applications, fixed datasets, or proprietary hardware

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stacks, making it difficult to compare runtime efficiency or reproduce experimental performance across heterogeneous architectures [1, 11, 12].

This preliminary work addresses these technical gaps by introducing **EdgeBench**, a lightweight preprocessing and streaming ingestion framework engineered to normalize high-throughput data pipelines before execution by resource-intensive downstream anomaly detection systems. EdgeBench provides a modular, bounded-state streaming architecture that decouples source data acquisition from application logic, unifying diverse data inputs through a common interface while integrating low-overhead preprocessing, runtime telemetry collection, and hardware deployment into a reusable execution pipeline. The framework supports plug-and-play AI workloads without modifying the underlying infrastructure and automatically captures standardized system-level metrics across the edge-to-cluster continuum.

The core EdgeBench framework coordinates five primary pipeline blocks through a standardized, uniform payload contract. First, the *Unified Ingestion Layer* abstracts raw data inputs, handling finite validation records and unbounded, real-time sensor streams identically by decoding and piping exactly one frame per loop iteration. Second, the *Streaming Feature Preprocessing* stage converts incoming frames to single-channel grayscale matrices and calculates the mean absolute inter-frame difference to isolate temporal variations without look-ahead dependencies. Third, a *Bounded State Buffer* limits system context to an active historical window, automatically discarding stale payloads to maintain a flat physical memory footprint independent of stream duration. Fourth, an *Interchangeable Inference Interface* exposes a uniform payload contract supporting three plug-and-play detection configurations: a rule-based *Threshold Detector*, a pretrained *ML Detector*, and a concurrent *Hybrid Detector*. Finally, a non-intrusive *Telemetry Monitoring Engine* samples the physical hardware to compile standardized metrics, including frame processing throughput (FPS), end-to-end latency, CPU core loads, and independent inference execution times.

Due to constrained development timelines, this **initial validation** evaluates EdgeBench as a streaming video preprocessing tier on both an x86 desktop workstation (representing a high-compute aggregation node) and a low-power, embedded ARM-based Xilinx PYNQ platform (representing an instrument-adjacent compute node). Video workloads serve as a high-bitrate proxy for intense edge data ingestion processing. Both platforms were stressed using a representative real-time anomaly detection application fed by identical high-bitrate prerecorded benchmarks and continuous live USB camera streams [13, 14]. The internal telemetry module automatically recorded hardware-level execution signatures across all pipeline states.

The preliminary experiments validate EdgeBench’s cross-platform portability. The x86 workstation profiles unconstrained processing (MP4) at 0.7583 ms latency and 574.2381 FPS (0.9543 MB/s), requiring 0.6051 ms (capture), 0.6059 ms (preprocessing), 0.1202 ms (inference), and 0.3561 ms (logging). Under live constraints (Camera), latency scales to 1.7610 ms (29.7223 FPS, 26.1231 MB/s) due to a 31.3686 ms capture loop, while preprocessing (1.5326 ms), inference (0.2073 ms), and logging (0.5033 ms) remain minimal. Conversely, the ARM PYNQ platform reveals critical front-end bottlenecks. Live streaming increases latency to 29.9805 ms, throttling throughput to 19.7063 FPS (17.3200 MB/s). Telemetry isolates the delay within ingestion layers: hardware capture requires 19.2213 ms and preprocessing takes 28.7432 ms, eclipsing inference (1.1553 ms) and logging (1.4280 ms). Similarly, embedded MP4 processing yields 8.2160 ms latency (66.3549 FPS, 0.1103 MB/s), driven by capture (5.4820 ms) and preprocessing (7.2289 ms) over inference (0.8196 ms) and logging (1.2806 ms). This justifies using low-overhead ingestion governors to absorb front-end costs before data hits heavy downstream classifiers. Furthermore, runtime telemetry confirms a completely flat physical memory footprint independent of stream duration. Peak Resident Set Size (RSS) remains stable at 197.2617 MB (Camera) and 219.7773 MB (MP4) on the x86 node, and 143.0898 MB (Camera) and 147.4297 MB (MP4) on the embedded platform.

Future work will integrate physical interfaces for multi-modal industrial sensor arrays (vibration, thermal, acoustic) to expand validation beyond video proxies, establishing a scalable foundation to balance localized edge preprocessing with high-performance computing backends.

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